

VAPO

Value-Model-Based RL for Advanced Reasoning

60.4

AIME 2024

avg@32 • Surpasses DAPO
by 10+ pts

Base Model: Qwen2.5-32B • No SFT Data Required

ByteDance Seed

Why Value Models?

Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO/DAPO)

- Trajectory-level advantage
- Uniform credit assignment across all tokens
- Subtle step errors cause catastrophic failures
- Limited precision for multi-step reasoning

Value-Model-Based (VAPO)

- Token-level credit assignment
- Traces each action's impact on returns
- Lower variance estimates per token
- Generalizes across samples

Value models enable precise token-level credit assignment — critical for complex reasoning

Three Challenges in Value-Model-Based RL

Long chain of thought tasks expose fundamental limitations

1

Value Model Bias

Long trajectories compound estimation errors through bootstrapping. The value model becomes increasingly biased as horizon extends.

2

Heterogeneous Lengths

Different response lengths require different bias-variance tradeoffs. Short responses need lower variance; long responses need lower bias.

3

Reward Sparsity

Only the final <eos> token receives reward signal. Long reasoning chains suffer from extremely sparse feedback.

VAPo Framework

Integrated Solutions for Value-Model-Based Reasoning RL

Value-Pretraining

Pre-trains value model before RL to reduce initial bias

Decoupled GAE

Separates GAE computation for robust value estimation (from VC-PPO)

Length-Adaptive GAE

Dynamic λ adjustment:
short $\rightarrow$ low λ (low variance),
long $\rightarrow$ high λ (low bias)

Clip-Higher

Asymmetric clipping for exploration (from DAPO)

Token-level Loss

Normalizes loss at token level vs sequence level (from DAPO)

Positive Example LM Loss

Self-imitation learning on correct trajectories

Group-Sampling

Multiple responses per prompt for better baseline (from GRPO)

Challenge Mapping:



Bias



Length



Reward



Implicit

Length-Adaptive GAE

Dynamic λ Adjustment for Variable Response Lengths

Core Insight

The λ parameter in GAE controls the bias-variance tradeoff: low λ reduces variance but increases bias, while high λ reduces bias but increases variance. Different response lengths need different tradeoffs.

Short Responses

λ Lower

- Shorter horizon
- Less bootstrapping error
- Prioritize variance reduction

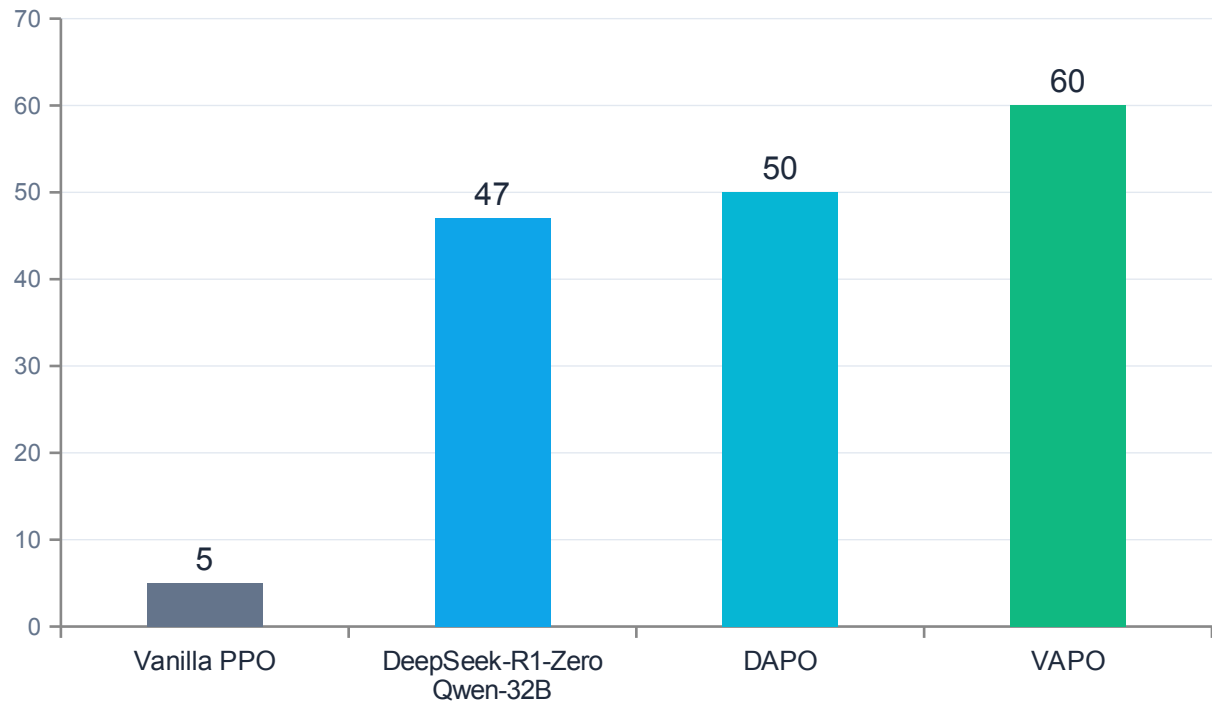
Long Responses

λ Higher

- Longer horizon
- More bootstrapping error accumulates
- Prioritize bias reduction

VAPO Scores 60 on AIME 2024

10+ Points Above DAPO and DeepSeek-R1-Zero



+49 pts

vs Vanilla PPO

+10 pts

vs DAPO

Training Efficiency

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO

~5K

Steps to SOTA

60%

Peak Accuracy

0

Training Crashes

~6K

Avg Response Tokens

Training Curve Insights (figure_2.jpg)

- VAPO consistently outperforms DAPO throughout training
- VAPO reaches 60% accuracy around step 4,500
- DAPO plateaus near 50% at step 5,500
- DeepSeek-R1-Zero-Qwen-32B reaches 47% after ~10,000 steps

Response Length (figure_3.jpg):

VAPO generates longer reasoning chains than DAPO (~6,000 vs ~4,500 tokens), suggesting more elaborate reasoning processes.

Ablation: Every Component Matters

Value-Pretraining Alone Adds 49 Points

Component Removed	AIME 2024	Drop
VAPO (Full)	60	—
w/o Value-Pretraining	11	-49
w/o Decoupled-GAE	33	-27
w/o Length-adaptive GAE	45	-15
w/o Clip-Higher	46	-14
w/o Token-level Loss	53	-7
w/o Positive Example LM Loss	54	-6
w/o Group-Sampling	55	-5

Critical Finding: Value-Pretraining is the single most impactful component. Without it, value-model-based RL barely works (drops from 60 to 11).

Key Takeaways

- 1 Value-model-based RL can surpass value-model-free methods when core challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed.
- 2 Length-adaptive GAE is a novel technique that dynamically optimizes the bias-variance tradeoff across heterogeneous response lengths.
- 3 Value-Pretraining is the single most important component — without it, value-model-based RL barely works (49-point drop from 60 to 11).
- 4 VAPO is highly training-efficient and stable, reaching SOTA in ~5,000 steps with zero training crashes across multiple runs.
- 5 Implications: The path to better reasoning models lies in sophisticated credit assignment, not just more data or larger models.